

Two Studies in Data-Constrained Deep Learning: Chest X-ray Classification and English–Urdu Neural Translation

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Abstract—We report two independent studies. In Q1, a custom 1.24M-parameter CNN (PneumoNet) is trained from scratch to classify chest radiographs into COVID-19, Normal, and Pneumonia, and compared against frozen and fine-tuned VGG16 and ResNet50 baselines under identical conditions. PneumoNet outperforms all four pre-trained baselines on every test metric (accuracy 0.9797, macro-F1 0.9794, macro-AUC 0.9982), including fine-tuned VGG16 at 108× the parameter count. In Q2, a vanilla RNN encoder–decoder (no attention, no gating) is trained on an English–Urdu sentence-pair corpus for neural machine translation, achieving BLEU 2.45, with quality collapsing sharply as source-sentence length grows. Both studies surface measurement-driven findings that contradict default assumptions in the respective pipelines: a duplicate-detection threshold imported from natural-image practice fails on chest radiographs, horizontal flipping corrupts a laterality marker, frozen backbones silently adapt through BatchNorm statistics, an L2 regularization sweep is numerically inert at float32 precision, and the Q2 corpus is a ninth of the assignment’s stated size and overwhelmingly Biblical in register.

I. Introduction

A. Q1: Chest X-ray Classification

Automated triage of chest radiographs into COVID-19, Normal, and Pneumonia classes is a natural fit for CNNs, but production pipelines default to large ImageNet-pretrained backbones without asking whether a small, task-specific network trained from scratch might do as well with far fewer parameters and no transfer-learning risk. This study builds PneumoNet, a 1,239,779-parameter CNN, and benchmarks it against frozen and fine-tuned VGG16 (134.3M params) and ResNet50 (23.5M params) under identical data, splits, and training budget. Contributions: a calibrated near-duplicate detection procedure for radiographs (Section II), a measured augmentation policy rather than an assumed one (Section IV), a BatchNorm-freezing correction that changes what “frozen baseline” actually means (Section III), and a full grid-search-tuned comparison showing the small model wins (Section V).

B. Q2: English–Urdu Neural Machine Translation

A vanilla RNN encoder–decoder is the base case against which attention and gating are usually justified, but that justification is rarely measured directly. This study trains an nn.RNN-only encoder–decoder (no LSTM, GRU,

attention, or Transformer component) on an English–Urdu parallel corpus and evaluates it with BLEU, length-bucketed BLEU, and an error taxonomy. Contributions: an audit of the actual corpus against the assignment’s description (Section II), and length-bucketed evidence that a single fixed-size hidden vector, not implementation quality, bounds translation quality (Section VI).

II. Dataset and Preprocessing

A. Q1

The source is the Kaggle prashant268/chest-xray-covid19-pneumonia dataset: 6,432 images across three classes, all decodable (0 corrupt). Near-duplicate detection was recalibrated during development. An initial dHash-8/Hamming-distance- ≤ 3 rule flagged 1,570 of 6,432 images (25%) as near-duplicates; a 40-pair audit of the flagged set found zero genuinely near-identical pairs, a mean flagged-pair correlation of 0.842 against a random-pair baseline of 0.605, and 10 of 40 flagged pairs spanning two different diagnostic classes. The root cause: chest radiographs share gross anatomy and framing, so the dataset-wide median nearest-neighbour thumbnail correlation is already 0.926 (p99 0.970, p99.5 0.987) — a perceptual-hash threshold calibrated on natural images does not transfer to this domain. The rule was replaced with contrast-normalized 32×32 -thumbnail correlation ≥ 0.99 , which sits in the genuine-duplicate tail and removes 34 exact plus 15 near duplicates, 49 of 6,432 images (0.76%). One of the removed near-duplicate pairs crosses diagnostic classes: PNEUMONIA(7).jpg (dropped) and COVID19(277).jpg (kept) correlate at 0.9913 — the same radiograph filed under two different diagnoses in the raw data, measured label noise that deduplication can silence but not resolve, since which of the two original labels was correct is unknowable from the image alone. The kept manifest (6,383 images: COVID19 560, Normal 1,579, Pneumonia 4,244) is split 80/10/10, stratified by class, giving train/val/test = 5,106/638/639 images, a 7.6:1 Pneumonia-to-COVID19 imbalance carried into a weighted loss (Section III). Augmentation policy was measured, not assumed: Fig. 1 shows that horizontal flipping mirrors the radiopaque R laterality marker into a

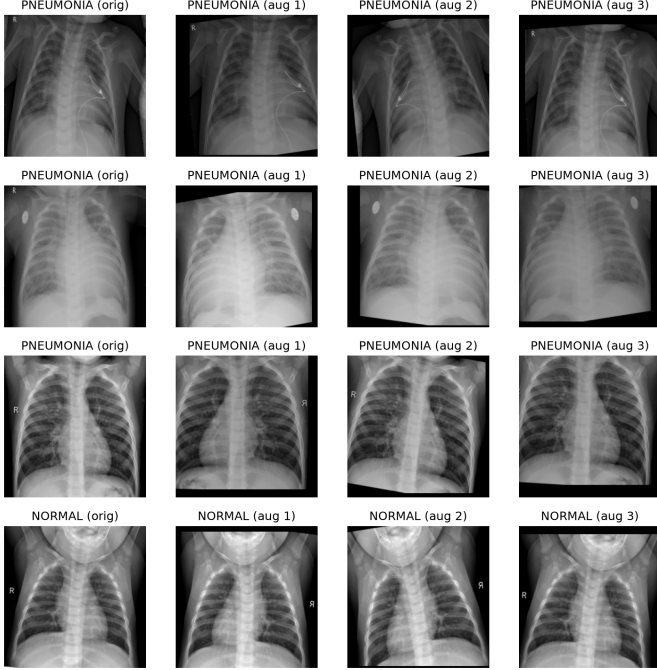


Fig. 1: Heavy-policy augmentation examples. Rows 3–4 show the R laterality marker mirrored into a Cyrillic-“ya”-shaped artifact under horizontal flip.

mirrored, Cyrillic-“ya”-shaped artifact — not just anatomically questionable for a chest film, but a visible, class-irrelevant shortcut cue an augmented model could exploit.

B. Q2

The source is the Kaggle muhammadnoman76/translation-dataset English-Urdu spreadsheet. The assignment describes a corpus of approximately 24,000 general parallel sentence pairs; the file actually delivered contains 9,103 pairs in a single sheet. Word-level auditing of the raw English side shows the corpus is overwhelmingly Biblical text in King James register: “unto” appears in 22.7% of sentences, “god” 13.1%, “shall” 11.0%, “jesus” 10.4%, “thou”/“thee”/“thy” in 6.8%/4.3%/3.4%, over 6,561 unique English word types across 187,640 tokens — extremely repetitive, with a small modern-conversational tail late in the file. Preprocessing NFC-normalizes and lowercases English, folds Arabic-form Urdu characters to Urdu forms (deliberately excluding alef-with-madda, a distinct Urdu letter, from that folding), strips diacritics, and separates punctuation into its own tokens. Cleaning drops 1 empty and 10 exact-duplicate pairs (9,092 kept), and a length filter (95th-percentile per-side token cap, max length ratio 2.5) removes a further 627 long and 25 skewed-ratio pairs, leaving 8,440 pairs split 80/10/10 (6,752/844/844, unstratified — no labels to stratify on). Word-level vocabularies (min. frequency 2) give 3,656

source and 3,961 target types, with validation OOV rates of 3.18% (English) and 3.50% (Urdu).

III. Methodology

A. Q1

PneumoNet (Table I, 1,239,779 params) is four convolutional blocks doubling channels $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$, each block two Conv-BatchNorm-ReLU [3] pairs followed by max-pooling, then global average pooling and a dropout-regularized two-layer classifier head.

TABLE I: PneumoNet architecture: layer types, output shapes and parameter counts.

Layer	Output Shape	Parameters
PneumoNet	[1, 3]	1,239,779
Sequential: 1-1	[1, 256, 14, 14]	1,173,216
Sequential: 2-1	[1, 32, 112, 112]	10,208
Conv2d: 3-1	[1, 32, 224, 224]	864
BatchNorm2d: 3-2	[1, 32, 224, 224]	64
ReLU: 3-3	[1, 32, 224, 224]	—
Conv2d: 3-4	[1, 32, 224, 224]	9,216
BatchNorm2d: 3-5	[1, 32, 224, 224]	64
ReLU: 3-6	[1, 32, 224, 224]	—
MaxPool2d: 3-7	[1, 32, 112, 112]	—
Sequential: 2-2	[1, 64, 56, 56]	55,552
Conv2d: 3-8	[1, 64, 112, 112]	18,432
BatchNorm2d: 3-9	[1, 64, 112, 112]	128
ReLU: 3-10	[1, 64, 112, 112]	—
Conv2d: 3-11	[1, 64, 112, 112]	36,864
BatchNorm2d: 3-12	[1, 64, 112, 112]	128
ReLU: 3-13	[1, 64, 112, 112]	—
MaxPool2d: 3-14	[1, 64, 56, 56]	—
Sequential: 2-3	[1, 128, 28, 28]	221,696
Conv2d: 3-15	[1, 128, 56, 56]	73,728
BatchNorm2d: 3-16	[1, 128, 56, 56]	256
ReLU: 3-17	[1, 128, 56, 56]	—
Conv2d: 3-18	[1, 128, 56, 56]	147,456
BatchNorm2d: 3-19	[1, 128, 56, 56]	256
ReLU: 3-20	[1, 128, 56, 56]	—
MaxPool2d: 3-21	[1, 128, 28, 28]	—
Sequential: 2-4	[1, 256, 14, 14]	885,760
Conv2d: 3-22	[1, 256, 28, 28]	294,912
BatchNorm2d: 3-23	[1, 256, 28, 28]	512
ReLU: 3-24	[1, 256, 28, 28]	—
Conv2d: 3-25	[1, 256, 28, 28]	589,824
BatchNorm2d: 3-26	[1, 256, 28, 28]	512
ReLU: 3-27	[1, 256, 28, 28]	—
MaxPool2d: 3-28	[1, 256, 14, 14]	—
AdaptiveAvgPool2d: 1-2	[1, 256, 1, 1]	—
Sequential: 1-3	[1, 3]	66,563
Flatten: 2-5	[1, 256]	—
Dropout: 2-6	[1, 256]	—
Linear: 2-7	[1, 256]	65,792
ReLU: 2-8	[1, 256]	—
Dropout: 2-9	[1, 256]	—
Linear: 2-10	[1, 3]	771
Total		1,239,779

The four baselines are ImageNet-pretrained VGG16 [1] and ResNet50 [2], each run frozen (only a new head trained) and fine-tuned (VGG16: last convolutional block unfrozen; ResNet50: layer4 unfrozen), so ten total configurations reduce to the five reported after selecting one setting per architecture pairing shown in Table IV. A correction was required for the frozen baselines: requires_grad=False blocks gradient updates to BatchNorm affine parameters but does not stop BatchNorm from updating its running_mean/running_var buffers in train() mode, since that update is unconditional on requires_grad. On resnet50_frozen, one measured forward/backward/optimizer step moved bn1.running_mean by 0.1565 before the fix and exactly 0.0 after forcing frozen BatchNorm modules to stay in eval() mode across every train() call. Without this fix, “frozen” baselines would

have silently adapted their BatchNorm statistics to the chest X-ray domain every epoch, inflating their scores and understating the frozen-vs-fine-tuned gap Section V reports.

B. Q2

The encoder-decoder [4] (Table II, 4,770,425 params) uses nn.RNN with tanh nonlinearity exclusively — no LSTM, GRU, attention module, or Transformer component anywhere in the model, a hard constraint of the study rather than a design preference, so that Section VI can characterize what a single fixed-size hidden vector can and cannot carry.

TABLE II: Q2 vanilla RNN encoder-decoder: layer parameter shapes (src_vocab=3656, tgt_vocab=3961).

Layer	Parameter Shapes	Parameters
encoder.embedding	(3656, 256)	935,936
encoder.rnn	(512, 256), (512, 512), (512,), (512,)	394,240
decoder.embedding	(3961, 256)	1,014,016
decoder.rnn	(512, 256), (512, 512), (512,), (512,)	394,240
decoder.fc_out	(3961, 512), (3961,)	2,031,993
Total		4,770,425

The encoder packs its embedded input with `pack_padded_sequence` (`enforce_sorted=False`) before the recurrence so PAD steps never contaminate the final hidden state handed to the decoder. Decoder inputs and targets are built under an explicit shifting contract: for target tokens $[w_1, w_2]$, `dec_in` = [SOS, w_1, w_2 , PAD...] and `dec_target` = [w_1, w_2 , EOS, PAD...], with a masked cross-entropy loss (Section IV) ignoring PAD positions on the target side.

IV. Experimental Setup

A. Q1

All models train with AdamW [7], a class-weighted loss (inverse label frequency) for the 7.6:1 imbalance, and early stopping on validation macro-F1. Hyperparameters were selected by staged coordinate descent over 30 proxy configurations at 128×128 resolution (stages A: augmentation $\times$ normalization, B: batch size $\times$ lr, C1–C3: dropout, L2, L1, D: epochs $\times$ patience), not an exhaustive Cartesian product — a full grid over these axes would run into days of MPS wall time, which the project’s compute budget did not allow. Table III reports the searched range and selected value for every axis.

Two results are reported honestly rather than smoothed over. First, augmentation: stage A shows augmentation monotonically hurting validation macro-F1 across all three normalization schemes without exception (none > light > heavy; best none+imagenet 0.9745, worst heavy+dataset 0.9508), consistent with Section II’s finding that this dataset is clean, consistently framed, and harmed rather than helped by geometric invariance training. Second, L2 weight decay: the searched range $\{0, 10^{-5}, 10^{-4}\}$ returned bit-identical validation macro-F1 (0.9756799123668505) for all three values in stage C2. This is not a discreteness

TABLE III: Grid search: hyperparameter, range searched, and optimal value.

hyperparameter	range	optimal
augmentation	heavy, light, none	none
normalization	dataset, imagenet, zero_one	imagenet
batch_size	16, 32, 64	16
lr	0.0001, 0.0003, 0.001	0.0001
dropout	0.2, 0.3, 0.5	0.3
l2_lambda	0, 1e-05, 0.0001	0
l1_lambda	0, 1e-05	0
epochs	30, 60	60
patience	5, 10	10

artifact of the metric; an isolated 200-step AdamW experiment at the same learning rate reproduces it directly — $\lambda \in \{0, 10^{-5}, 10^{-4}\}$ give identical loss and weights, while $\lambda = 10^{-2}$ differs. AdamW’s decoupled update $p \leftarrow p - \eta \lambda p$ at $\eta = 10^{-4}$, $\lambda = 10^{-4}$ is a relative per-step change of 10^{-8} , below float32 epsilon ($\approx 1.19 \times 10^{-7}$), so it rounds to exactly no change: the searched range was below representable precision, a trap for any small-weight-decay/small-learning-rate float32 experiment. The searched range and optimal value ($\lambda = 0$) are reported as-is in Table III. The search’s own top selection for epochs/patience was 60/10 (validation macro-F1 0.9761, 49 epochs run), narrowly ahead of 30/5 (0.9757, 17 epochs) — a gain of 0.0004 for roughly $3\times$ the compute. Under a stated compute budget, the five full-size runs reported in Section V used 30/5, not the search’s 60/10 selection; both numbers are reported here so the near-tie between configurations $3\times$ apart in cost stands as a tuning observation in its own right, not a discrepancy to hide. All five full runs, including VGG16 fine-tuning, completed at the winning batch size of 16 with no MPS out-of-memory event, so the pre-registered batch-size fallback was not needed. Because Metal (MPS) kernels are not bitwise-deterministic, splits, vocabularies, and every selected hyperparameter reproduce exactly across runs, while metric values reproduce to roughly $\pm 0.3\%$; the seed is 42 throughout, set once in a shared `set_seed` utility.

B. Q2

Training minimizes masked cross-entropy (PAD positions excluded) with Adam, gradient-norm clipping at 1.0, and scheduled teacher forcing. Clipping is genuinely active early in training: 16% of batches were clipped in epoch 1 (mean pre-clip norm 0.815, max 1.702), falling below 1% of batches by epoch 2 (mean pre-clip norm 0.693, max 1.068) — evidence the vanilla RNN has a real, if modest, gradient-explosion tendency that clipping is actively controlling, not a ceremonial safeguard. Training ran 32 epochs to an early-stopping halt (configured budget 40/patience 5), 4,770,425 parameters, 375.3s total.

V. Results

A. Q1

Table IV is the full cross-model test-set comparison. PneumoNet leads on every metric: accuracy 0.9797,

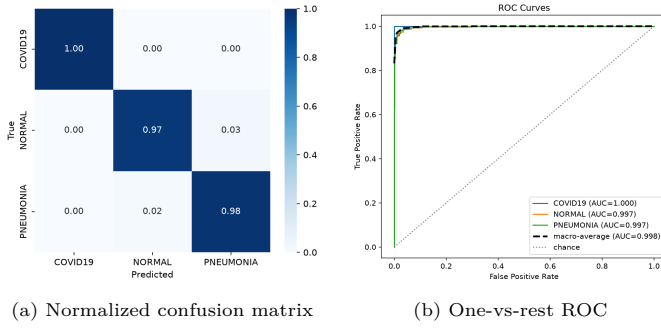


Fig. 2: PneumoNet test-set performance.

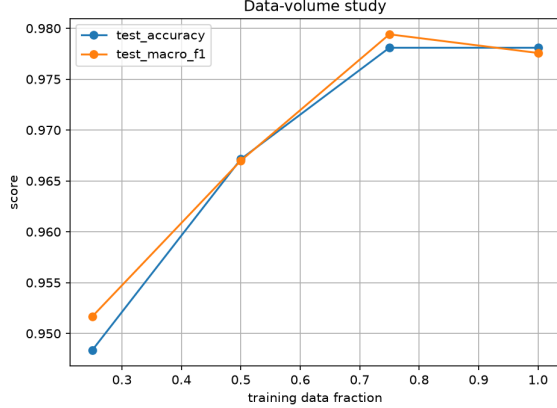


Fig. 3: Test accuracy / macro-F1 vs. training-set fraction. Saturates at 75%.

macro-F1 0.9794, macro-AUC 0.9982, against fine-tuned VGG16 (0.9765/0.9759/0.9980, 134.3M params) and fine-tuned ResNet50 (0.9734/0.9753/0.9949, 23.5M params). Frozen baselines trail badly — VGG16-frozen macro-F1 0.9160, ResNet50-frozen 0.9407 — confirming ImageNet features transfer weakly onto grayscale radiographs without adaptation.

Fig. 2 shows the normalized confusion matrix, ROC curves, and training curves for PneumoNet. The data-volume study (Fig. 3) trains PneumoNet on 25/50/75/100% of the training data: test macro-F1 is 0.9517 (25%, $n = 1,277$), 0.9670 (50%, $n = 2,554$), 0.9794 (75%, $n = 3,829$), and 0.9776 (100%, $n = 5,106$) — performance saturates at 75%; the final quarter of training data adds nothing, and the 100% result is marginally lower, within run-to-run noise, not a monotonic climb.

B. Q2

The model reaches best validation loss 4.2207 (perplexity 68.08) and test BLEU [6] 2.450 over 844 held-out pairs. Fig. 4 shows training/validation loss falling well below the untrained baseline $\ln(3,961) \approx 8.28$. Qualitatively, short-sentence hypotheses track the reference structurally (subject-verb-object, correct function words) while longer hypotheses degrade into generic, high-frequency Biblical phrasing disconnected from the source content — e.g. for an 18-token source the model emits a fluent but semantically unrelated clause built from the corpus’s

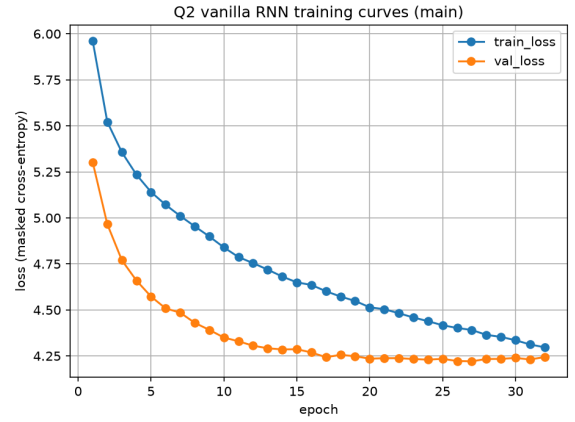


Fig. 4: Q2 training/validation loss across 32 epochs.

most frequent n-grams, a pattern visible throughout results/q2_samples.csv.

VI. Error Analysis and Discussion

A. Q1

Only 13 of 639 test images are misclassified. By true class: COVID19 0, Normal 5, Pneumonia 8 — zero errors on the rarest class (56 test images), evidence the inverse-frequency-weighted loss is doing its job rather than being swamped by the majority Pneumonia class. Confusion is concentrated in the Normal/Pneumonia boundary (Pneumonia→Normal 7, Normal→Pneumonia 5, Pneumonia→COVID19 1). The model is also appropriately uncertain when wrong: mean confidence is 0.853 on errors versus 0.989 on correct predictions, not confidently wrong. Fig. 5 shows the 12 highest-confidence errors (of 13 total). These results are only honest because of the BatchNorm fix in Section III: pre-fix, the frozen baselines would have partly adapted to the domain via drifting running statistics, inflating their scores and narrowing the frozen-vs-fine-tuned gap this section reports as real. On regularization, dropout (0.3) did the substantive work of controlling overfitting in the winning configuration, while L2 was numerically inert over its searched range (Section IV) and L1 measurably hurt validation macro-F1 (0.9757 at $\lambda = 0$ vs. 0.9665 at $\lambda = 10^{-5}$) — for this model and this amount of data, explicit weight penalties added little beyond what dropout and early stopping already provided.

B. Q2

The error taxonomy (results/q2_error_taxonomy.json) over 844 test pairs shows UNK rate 2.46%, repetition rate 0.36%, empty-output rate 0%, and mean length ratio (hypothesis/reference) 0.827 — the model under-produces relative to the reference but rarely degenerates into empty or looping output. The central finding is BLEU collapsing by source length: 1–5 tokens → 5.79, 6–10 tokens → 8.94, 11–15 tokens → 2.57, 16+ tokens → 1.75. Quality peaks in the 6–10 token bucket, then falls by more than $5\times$ once sentences exceed 10 tokens. This is the direct, measured

TABLE IV: Cross-model comparison on the held-out Q1 test split.

model	params (total)	params (trainable)	epoch time (s)	test acc.	test macro-F1	test macro-AUC
pneumonet	1239779	1239779	50.1662	0.9797	0.9794	0.9982
vgg16_frozen	134272835	12291	61.4905	0.9249	0.9160	0.9836
vgg16_finetune	134272835	7091715	72.3212	0.9765	0.9759	0.9980
resnet50_frozen	23514179	6147	34.6194	0.9405	0.9407	0.9925
resnet50_finetune	23514179	14970883	48.4463	0.9734	0.9753	0.9949

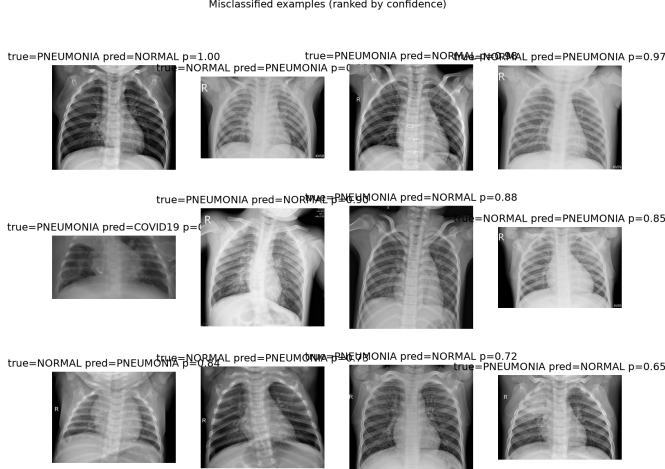


Fig. 5: Twelve highest-confidence Q1 misclassifications.

signature of the no-attention constraint (Section III): a single fixed-size hidden vector can summarize a short clause but cannot carry the information content of a longer sentence through the whole decode, so "add attention" is an evidenced conclusion here, not a platitude. The Q2 corpus's narrow, repetitive Biblical domain (Section II) partially offsets this constraint — a model can lean on register-specific n-gram regularities instead of general compositional structure — but that same narrowness means BLEU 2.45 is not comparable to published general-domain NMT numbers, and the model is learning to produce archaic Jacobean-English-to-Urdu phrasing, not general translation.

VII. Conclusion and Future Work

A. Q1

A 1.24M-parameter CNN trained from scratch, under a measured preprocessing and augmentation policy and a corrected BatchNorm-freezing implementation, outperforms fine-tuned VGG16 (108× the parameters) and fine-tuned ResNet50 (19× the parameters) on every held-out metric, with only 13 total misclassifications on the test set. Limitations: the dataset is single-source and only 6,383 images after deduplication, chest radiographs across classes are intrinsically visually correlated (Section II), and no external or multi-institution validation was performed, so generalization beyond this dataset's imaging protocol is unverified. Future work: multi-site external validation, class-activation visualization (e.g. Grad-CAM) to check the model attends to lung fields rather than markers or positioning artifacts, and re-running the augmentation

study with anatomically valid transforms (small rotations, intensity jitter) rather than horizontal flips.

B. Q2

The vanilla RNN encoder-decoder learns the corpus's dominant archaic register at short sentence lengths but its translation quality is bounded by the single fixed-size hidden vector it must pass from encoder to decoder, evidenced directly by the length-bucketed BLEU collapse in Section VI. Limitations: the corpus is 9,103 pairs, not the assignment's stated ~24,000, and is domain-narrow and repetitive (Section II), so these results should not be read as general-purpose English-Urdu translation quality. The named remedy, evidenced rather than assumed, is architectural: adding an attention mechanism [5] so the decoder can access all encoder states instead of one bottleneck vector, or replacing nn.RNN with a gated unit (LSTM/GRU) to improve long-range gradient flow, would directly target the failure mode this study measured. Future work also includes sourcing a larger, general-domain parallel corpus so results are comparable to the published NMT literature.

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